KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6342A | 6 Credits: 4 | Open Elective | Source-grounded

Introduction to Artificial Intelligence

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6342A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6342A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Artificial Intelligence & Machine Learning
Course code	6342A
Course title	Introduction to Artificial Intelligence
Semester	6 Credits: 4
Course category	Open Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

19 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	Explain the concept of Artificial Intelligence 13 Understanding Apply Mathematical logic for knowledge	See official syllabus
CO2	representation of AI 15 Understanding	See official syllabus
CO3	Demonstrate supervised machine learning techniques 15 Understanding	See official syllabus
CO4	Demonstrate Unsupervised Learning techniques 15 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the concept of Artificial Intelligence	5	As specified
Module 2	Apply Mathematical logic for knowledge representation of AI	6	As specified
Module 3	Demonstrate supervised machine learning techniques	5	As specified
Module 4	Demonstrate Unsupervised Learning techniques	3	As specified

MODULE 1

Explain the concept of Artificial Intelligence

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the concept of Artificial Intelligence
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

Define AI, foundation and history of AI. 2 Understanding Understanding

Define AI, foundation and history of AI. 2 Understanding Understanding is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define AI, foundation and history of AI. 2 Understanding Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define ai, foundation and history of ai. 2 understanding understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define AI, foundation and history of AI. 2 Understanding Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Define AI, foundation and history of AI. 2 Understanding Understanding □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Discuss applications of AI. 2

Discuss applications of AI. 2 is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss applications of AI. 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss applications of ai. 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss applications of AI. 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Discuss applications of AI. 2 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the Turing test.

Explain the Turing test. is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Turing test. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the turing test. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Turing test. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the Turing test. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain AI problems

Explain AI problems is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain AI problems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain ai problems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain AI problems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain AI problems [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.

Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. is an official topic in Module 1 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain agents and environments. 2 understanding contents : introduction to ai-ai concepts, foundations of ai, applications of ai -language models -

information retrieval - information extraction - natural language processing - machine translation - speech recognition, turing test. ai problems: water jug problem, tower of hanoi problem, four-queens problem, travelling salesman problem intelligent agents-agents and environments, structure of agent - types of agents. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain agents and environments. 2 Understanding Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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Explain the concept of Artificial Intelligence

  M1.01      Define AI, foundation and history of AI.               2
Understanding

Understanding
  M1.02      Discuss applications of AI.                            2

  M1.03      Explain the Turing test.                               3
Understanding

  M1.04      Explain AI problems                                    3
Understanding

  M1.05      Explain agents and environments.                       2
Understanding
Contents :
Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language
models
- Information retrieval - Information extraction - Natural language processing -
Machine
translation - Speech recognition, Turing test.
AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem,
Travelling Salesman problem
Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Apply Mathematical logic for knowledge representation of AI

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply Mathematical logic for knowledge representation of AI
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Understanding knowledge representation and reasoning Describe the representation and reasoning

2 Understanding knowledge representation and reasoning Describe the representation and reasoning is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding knowledge representation and reasoning Describe the representation and reasoning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding knowledge representation and reasoning describe the representation and reasoning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding knowledge representation and reasoning Describe the representation and reasoning means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding knowledge representation and reasoning Describe the representation and reasoning □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

about objects, relations, events, actions, time 2 Understanding and space

about objects, relations, events, actions, time 2 Understanding and space is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify about objects, relations, events, actions, time 2 Understanding and space using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, about objects, relations, events, actions, time 2 understanding and space should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What about objects, relations, events, actions, time 2 Understanding and space means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ about objects, relations, events, actions, time 2 Understanding and space □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Demonstrate Propositional Logic

Demonstrate Propositional Logic is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate Propositional Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate propositional logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate Propositional Logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Demonstrate Propositional Logic [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Demonstrate First Order Logic

Demonstrate First Order Logic is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate First Order Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate first order logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate First Order Logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Demonstrate First Order Logic [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain Soundness and Completeness

Explain Soundness and Completeness is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Soundness and Completeness using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain soundness and completeness should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Soundness and Completeness means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain Soundness and Completeness [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining.

Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. is an official topic in Module 2 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain forward and backward chaining 3 understanding knowledge representation and reasoning-ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. logic and inferences: propositional logic, first order logic, soundness and completeness, forward and backward chaining.

should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain Forward and Backward chaining 3 Understanding Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Apply Mathematical logic for knowledge representation of AI

             Summarize Ontologies, foundations                 of
  M2.01                                                             2
Understanding
             knowledge representation and reasoning
             Describe the representation and reasoning
  M2.02      about objects, relations, events, actions, time        2
Understanding
             and space

  M2.03      Demonstrate Propositional Logic                        3
Understanding

  M2.04      Demonstrate First Order Logic                          2
Understanding

  M2.05      Explain Soundness and Completeness                     3
Understanding

  M2.06      Explain Forward and Backward chaining                  3
Understanding
Contents:
Knowledge representation and reasoning-Ontologies,foundations of knowledge
representation      and       reasoning,      representing and reasoning about
objects,relations,events,actions, time and space.
Logic and Inferences: Propositional Logic, First Order Logic, Soundness and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Demonstrate supervised machine learning techniques

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate supervised machine learning techniques
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine

Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the machine learning concepts. 3 understanding describe the process involved in machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

3 Understanding learning techniques.

3 Understanding learning techniques. is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding learning techniques. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding learning techniques. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding learning techniques. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Understanding learning techniques. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe Supervised Learning

Describe Supervised Learning is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe Supervised Learning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe supervised learning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe Supervised Learning means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Describe Supervised Learning [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Demonstrate Classification techniques

Demonstrate Classification techniques is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate Classification techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate classification techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate Classification techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Demonstrate Classification techniques [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression

Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression is an official topic in Module 3 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate regression techniques 3 understanding machine learning - definition -applications - processes involved in machine learning, machine learning techniques-supervised learning, unsupervised learning and reinforcement learning, data preprocessing – normalization, missing value handling. supervised learning classification - learning a class from examples, logistic regression - k-nearest neighbours - support vector machine - naive bayes - decision tree - random forest regression- linear regression -multi linear regression should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Demonstrate Regression techniques 3 Understanding Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Demonstrate supervised machine learning techniques
  M3.01     Explain the Machine Learning concepts.                3
Understanding
            Describe the process involved in machine
  M3.02                                                           3
Understanding
            learning techniques.

  M3.03     Describe Supervised Learning                          3
Understanding

  M3.04     Demonstrate Classification techniques                 3
Understanding
  M3.05     Demonstrate Regression techniques                     3
Understanding

Contents:
Machine Learning - Definition -Applications - Processes involved in Machine
Learning,
Machine Learning Techniques-Supervised Learning, Unsupervised Learning and
Reinforcement Learning, Data preprocessing – Normalization, Missing value
handling.
Supervised Learning
Classification - Learning a Class from Examples, Logistic Regression - K-Nearest
Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest

Regression- Linear Regression -Multi Linear Regression
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Demonstrate Unsupervised Learning techniques

This module is presented from the official syllabus scope for Introduction to Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate Unsupervised Learning techniques
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Describe types of Clustering

Describe types of Clustering is an official topic in Module 4 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe types of Clustering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe types of clustering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe types of Clustering means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Describe types of Clustering □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Demonstrate Clustering techniques

Demonstrate Clustering techniques is an official topic in Module 4 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate Clustering techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate clustering techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate Clustering techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Demonstrate Clustering techniques [illegible] definition/meaning [illegible]. [illegible], steps, application [illegible]. Technical terms English-[illegible] [illegible].

Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 Introduction to Machine Learning - Course (nptel.ac.in)

Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 Introduction to Machine Learning - Course (nptel.ac.in) is an official topic in Module 4 of Introduction to Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 Introduction to Machine Learning - Course (nptel.ac.in) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate dimensionality reduction techniques 6 understanding clustering - discovering clusters - types of clustering: hierarchical, agglomerative clustering, divisive clustering, partitional clustering - k-means clustering, mean shift clustering dimensionality reduction – principal component analysis, factor analysis, multidimensional scaling, linear discriminant analysis. text / reference t/r book title/author stuart russell and peter norvig. artificial intelligence: a modern approach, 3rd t1 edition. prentice hall. t2 nilsson n.j., artificial intelligence - a new synthesis, harcourt asia pvt. ltd. ethem alpaydin, introduction to machine learning, 2nd edition, mit press 2010. t3 t4 tom mitchell, machine learning, mcgraw-hill, 1997 r1 elaine rich and kevin knight. artificial intelligence, tata mcgraw hill deepak khemani. a first course in artificial intelligence, mcgraw hill education r2 (india) online resources sl.no website link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 introduction to machine learning - course (nptel.ac.in) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial

Aspect	What to prepare
	Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 Introduction to Machine Learning - Course (nptel.ac.in) means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 Introduction to Machine Learning - Course (nptel.ac.in) □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Demonstrate Unsupervised Learning techniques

  M4.01     Describe types of Clustering                          3
Understanding

  M4.02     Demonstrate Clustering techniques                     6
Understanding

  M4.03     Illustrate Dimensionality Reduction techniques        6
Understanding
Contents:

Clustering - Discovering clusters - Types of Clustering: Hierarchical,
Agglomerative
Clustering, Divisive clustering, Partitional Clustering - K-means clustering,
Mean Shift
clustering

Dimensionality reduction – Principal Component               Analysis,   Factor
Analysis,
Multidimensional scaling, Linear Discriminant Analysis.
Text / Reference

 T/R                                    Book Title/Author
        Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern
Approach, 3rd
 T1
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define AI, foundation and history of AI. 2 Understanding Understanding. (3 marks)

Answer: Begin with the standard definition or identity of *Define AI, foundation and history of AI. 2 Understanding Understanding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Discuss applications of AI. 2. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss applications of AI. 2*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the Turing test.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the Turing test.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain AI problems. (3 marks)

Answer: Begin with the standard definition or identity of *Explain AI problems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Understanding knowledge representation and reasoning Describe the representation and reasoning. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding knowledge representation and reasoning Describe the representation and reasoning*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain about objects, relations, events, actions, time 2 Understanding and space. (3 marks)

Answer: Begin with the standard definition or identity of *about objects, relations, events, actions, time 2 Understanding and space*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Demonstrate Propositional Logic. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate Propositional Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Demonstrate First Order Logic. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate First Order Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding learning techniques.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding learning techniques.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe Supervised Learning. (3 marks)

Answer: Begin with the standard definition or identity of *Describe Supervised Learning*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Demonstrate Classification techniques. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate Classification techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 4

Define or explain Describe types of Clustering. (3 marks)

Answer: Begin with the standard definition or identity of *Describe types of Clustering*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Demonstrate Clustering techniques. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate Clustering techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 Introduction to Machine Learning - Course (nptel.ac.in). (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate Dimensionality Reduction techniques 6 Understanding Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis. Text / Reference T/R Book Title/Author Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. T2 Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010. T3 T4 Tom Mitchell, Machine Learning, McGraw-Hill, 1997 R1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education R2 (India) Online Resources Sl.No Website Link 1 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 2 Introduction to Machine Learning - Course (nptel.ac.in)*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define AI, foundation and history of AI. 2 Understanding Understanding**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding knowledge representation and reasoning Describe the representation and reasoning**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the Machine Learning concepts. 3 Understanding Describe the process involved in machine**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Describe types of Clustering**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.

Term	Meaning in this lesson
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://onlinecourses.swayam2.ac.in/aic20_sp06/preview
- Introduction to Machine Learning - Course (nptel.ac.in)

IN-PAGE SEARCH

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